MIL-I-3949D
31 December 1971
SUPERSEDING
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30 September 1965

MILITARY SPECIFICATION

INSIGNIA, RANK, OFFICER'S, METAL; GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers metal rank insignia worn by officers of the United States Armed Forces.

1.2 Classification.- Insignia shall be classified as specified on the applicable specification sheet (see 3.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
- FEDERAL
  - PPP-B-601 - Boxes, Wood, Cleated-Plywood
  - PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
  - PPP-B-636 - Box, Fiberboard
  - PPP-B-676 - Boxes, Set-Up
- MILITARY
  - MIL-F-495 - Finish, Chemical, Black for Copper Alloys

STANDARDS
- FEDERAL
  - Fed. Std. No. 151 - Metals; Test Method
- MILITARY
  - MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
  - MIL-STD-129 - Marking for Shipment and Storage

SPECIFICATION SHEETS
- MIL-I-3949/ - For applicable Military Specification Sheet see Supplement 1

DRAWINGS
- THE INSTITUTE OF HERALDRY
  - B-13-12 - Attaching Devices, Heraldic; Prong and Clutch Type

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(Copies of specifications, standards, specification sheets and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Figure 1 is a miniature reproduction of the referenced drawing and is attached for information purposes only.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

THE INSTITUTE OF HERALDRY, U. S. ARMY
Standard Color Chip Set

(A standard color chip may be obtained from the procuring activity.)

3. REQUIREMENTS

3.1 Detail requirements for individual insignia.- Detail requirements or exceptions applicable to a particular insignia shall be as specified on the specification sheet for that insignia. In the event of conflict between the requirements of this specification and any specification sheet referenced herein, the specification sheet shall govern.

3.1.1 Dimensions.- All dimensions given as reference dimensions are information for bidding purposes only. The actual dimensions will be controlled by the Government furnished hub.

3.2 Standard sample.- Unless otherwise specified (see 6.2), the contracting officer will furnish a standard sample which will be used for matching color and finish only. Other characteristics of the standard sample may vary from the requirements of this specification or applicable specification sheet.

3.3 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished insignia covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for examination and tests as specified in 4.2. The approval of the first article authorizes the commencement for production, but does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

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3.4 Materials.- Materials shall conform to the referenced specifications and the requirements specified herein.

3.4.1 Copper base alloys.- Copper base alloys shall be roll polished, free from pits, scale (including red oxide), dents, cracks, nicks, deep scratches, segregations and foreign inclusions that will not be removed in later processing. The chemical composition of the applicable copper base alloys shall be as shown in table I when tested as specified in 4.4.1.

TABLE I Chemical composition of copper base alloys.-

Alloy	Copper	Nickel 1/	Lead (Max)	Iron (Max)	Manganese (Max)	Zinc	Others (Max)
Red brass	84.0-86.0		.05	.05		Remainder	.15
Low brass	78.5-81.5		.05	.05		Remainder	.15
Yellow brass	64.0-68.5		.15	.05		Remainder	.15
Gilding metal	94.0-96.0		.03	.05		Remainder	.13
Nickel silver	63.0-66.5	16.5-19.5	.10	.25	.50	Remainder	

1/ Cobalt counting as nickel.

3.4.2 Silver.-

3.4.2.1 Sterling silver.- Sterling silver shall be not less than 92.5 percent pure silver.

3.4.2.2 Silver filled material.- When silver filled material is specified, the silver overlay shall be not less than 92.5 percent pure silver. The base of the silver filled material shall be any copper base alloy having a specific density of not less than .314 pounds per cubic inch and a copper content of not more than 96 percent pure copper.

3.4.3 Gold.-

3.4.3.1 Gold for plating.- Gold for plating shall be not less than 24 karat.

3.4.3.2 Gold filled material.- When gold filled material is specified, the gold shall be of the weight and karat specified on the applicable specification sheet.

3.4.4 Solder.- Solder shall be a silver alloy having a melting point of not less than 1075 degrees F.

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3.4.5 Lacquer.- Lacquer shall be a pale clear synthetic lacquer. The use of pigmented lacquer shall not be permitted.

3.4.6 Enamel.- Except for subdued insignia, enamel shall be vitreous type hard glass enamel pigmented to the approved shade of opacity or translucence. The color shall match the color of The Institute of Heraldry color chip specified and shall match the standard or approved sample. The enamel for subdued insignia shall be a flat synthetic baking enamel pigmented to the specified shade of brown, black or olive drab as applicable.

3.4.7 Aluminum.- Any suitable aluminum alloy may be used for annodized insignia (see 6.3).

3.5 Design.- The embossed design of all insignia shall be an exact replica of the design on the Government furnished hub (see 3.9) from which the contractor's working dies shall be extracted.

3.6 Construction.-

3.6.1 Stamping, trimming and piercing.- Insignia shall be struck in such a manner to insure a well defined die-struck edge. The insignia shall be trimmed and when applicable pierced to the die-struck outline. All edges shall be clean, free form burrs, drag, step and rough edges. The piercing and trimming operations shall not distort the design or alter the shape of the insignia.

3.6.2 Soldering or electronic fusion.- Unless otherwise specified, joints may be soldered or electronically fused. Soldered connections shall be completely joined without excess solder and all excess flux shall be removed. Joints shall be clean, strong, smooth and free from burned or reduced areas. When tested as specified in 4.4.7, the soldered connection shall not crack or separate. When tested as specified in 4.4.3, the prongs shall not separate from the insignia at the fused joint.

3.6.3 Forcers.- When forcers are used there shall be no sharp or rough edges or corners. The impression of the forcer shall be symmetrical upon the insignia back with sufficient metal left to insure clean, smooth trimming. Forcers shall not be used on miniature or enameled insignia.

3.7 Detail of components.-

3.7.1 Attaching devices.-

3.7.1.1 Prong and clutch.- Unless otherwise specified, the winged clutch and notched post shall be used on all insignia. Prong and clutch attaching device shall conform to drawing B-13-12. The prongs shall be located as specified

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on the applicable specification sheet. Prongs shall be made from nickel silver specified in 3.4.1. The clutches shall be made from any type brass. Unless otherwise specified on the applicable specification sheet, prongs shall be 5/16 plus or minus 1/64 inch in length and shall be soldered or fused to the back of the insignia. When tested as specified in 4.4.2, it shall not be possible to remove the clutches by hand from the prong without first depressing the clutch fastener release wings.

3.7.1.2 Registering prong.- When a registering prong is required, the prong shall be 1/8 inch, plus or minus 1/64 inch in length and 0.026 inch, plus or minus .002 inch in diameter. Unless otherwise specified, the prong shall be made of nickel silver and shall be secured to the insignia by soldering or electronic fusion.

3.8 Finish.- The front and sides of the insignia shall be finished as specified on the applicable specification sheet. When plating is specified the entire insignia excluding attaching devices shall be plated. The backs of gold plated insignia shall be bright and clean. The backs of rhodium plated insignia shall be clean. Except for Navy insignia, gold plated or gold filled insignia may be lacquered at the option of the manufacturer. All Navy and Marine Corps gold plated or gold filled insignia shall be lacquered. When an insignia is lacquered the entire insignia shall be coated with lacquer specified in 3.4.5. The lacquer film shall be hard, level, adherent, continuous and free from lint, dust or other inclusions. While it is necessary to allow a tolerance range for the finish, insignia packaged in pairs shall have compatible finishes. In no case will insignia with finishes at the opposite ends of the tolerance range be mounted on the same card.

3.8.1 Cutting down.- When necessary to remove nicks, scratches, inequalities, pores or other blemishes, the insignia shall be cut down by means of grease or oil buffing. The insignia shall not be cut down to the extent that any of the design is obliterated.

3.8.2 Plating.- All plating shall be accomplished by electroplating methods. The plating shall be continuous and unbroken over the entire plated surface. There shall be no cut-through, shaded, peeled or blistered plating. Electroplating of the attaching devices is not required.

3.8.2.1 Gold plating.- The gold specified in 3.4.3.1 shall be used for gold plating. Items to be plated shall be cleaned of all matter that may affect the gold plating. The use of white metals as an undercoating for gold shall not be permitted. No visible chemical reaction shall appear when the gold plating is tested as specified in 4.4.5.

3.8.2.2 Rhodium plating.- When tested as specified in 4.4.6, there shall be no perceptible change in brightness of front of the insignia, except that along the edges there may be a slight darkening.

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3.8.3 Enameling.- Except for subdued insignia, enameling shall be accomplished using hard enamel specified in 3.4.6. The enamel shall be uniform in color and shall be smooth, flush, stoned level with the metal dikes, felted and polished. The enamel shall be free from cracks, bubbles, pits and foreign inclusions. The face and edges of all subdued insignia shall be coated with enamel specified in 3.4.6 and baked. Prior to enameling the entire subdued insignia shall be given a black chemical finish conforming to MIL-F-495. When tested as specified in 4.4.4.1, the enamel film shall not flake or chip. When tested as specified in 4.4.4.2, there shall be no appreciable transfer of color to the cotton swab.

3.8.4 Anodizing.- All aluminum shall be anodized to a minimum depth of 0.0003 inch and dyed (when applicable) to the required color. The dyed insignia shall be sealed in a nickel acetate bath. When tested as specified in 4.4.8, no stain shall appear on the dyed surface.

3.9 Government loaned property.- Hubs shall be furnished by the Government and shall be used only for making contractor's working dies. The contractor's stamping and embossing dies shall be tooled and polished to remove dents, nicks, scratches or other imperfections and the unornamented sections shall be smooth and polished. The tooling and polishing shall not be done to the point where the edges or details of the design fail to be true impressions of the hub furnished by the U. S. Government. Hubs will be used for the making of the contractor's working dies necessary for one contract or order.

3.10 Weight.- Weight specified on the applicable specification sheet does not include clutches, but does include prongs. The weight shall be determined by weighing 12 insignia of one type and finding the average weight.

3.11 Marking.-

3.11.1 Marking for identification.- The contractor shall stamp his trademark or other identifying mark and the letters "G.I." (for Government procurement only) legibly and inconspicuously on the back of each insignia. In addition, precious metal insignia shall bear a quality marking. Silver filled insignia may be marked in any of the following ways: 1/20 SF, 1/20 Sil. Filled or 1/20 Silver Filled.

3.11.2 Marking for cleaning of insignia.- In addition to the contract marking requirements, each mounting card (see 5.1) shall have the following statement printed on the face of the card: TO CLEAN RUB GENTLY WITH A SOFT DRY CLOTH. DO NOT CLEAN WITH A CHEMICALLY TREATED CLOTH OR ABRASIVE.

3.12 Workmanship.- Insignia shall be clean, well made and shall conform to the quality levels established by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article.- When a first article is required, examination shall be made on a completely fabricated insignia for all provisions of this specification applicable to end product examination and tests.

4.3 Sampling for inspection and acceptance.- Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component testing.- In addition to the quality assurance provisions of the subsidiary specifications and drawings, testing shall be performed on components and materials listed in table II for the characteristics shown. The inspection level shall be S-1 and the requirements are applicable to the lot average.

TABLE II Component testing.-

Components and lots expressed in terms of	Characteristic	Rqm't ref	Test method	#determ per s/u	Results reported as	Sample unit
Copper base alloy (as applicable) (1 lb.)	Chemical composition	3.4.1	4.4.1	2 for each element	Nearest 0.1% for each element	4 oz. of material composite for each alloy

4.3.2 Certification of compliance.- Components and materials listed herein may be accepted on the basis of a contractor's certification of compliance for characteristics indicated.

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Components	Characteristic	Rqm't Ref
Sterling silver	Fineness	3.4.2.1
Gold filled material	Karat and lamination weight	3.4.3.2 and applicable specification sheet
Gold for plating	Karat and underplating	3.4.3.1 & 3.8.2.1
Silver filled material	Fineness of silver overlay, specific density of base material, percent of pure copper in base material	3.4.2.2 and applicable specification sheet
Aluminum	Material identification	3.4.7
Anodizing	Depth	3.8.4
Rhodium plating	Material identification	3.8.2.2
Solder	Material identification	3.4.4
Lacquer	Material identification	3.4.5
Enamel (hard)	Material identification and shade opacity	3.4.6 and applicable specification sheet
Enamel (baked)	Material identification and shade	3.4.6 and applicable specification sheet
Attaching device (as applicable)	Material identification	3.7.1.1, 3.7.1.2 and applicable specification sheet

4.3.3 In-process inspection.- Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemlies are accomplished as specified. Whenever a deviation is noted, correction shall be made. Failure to make immediate correction may cause rejection of the affect end item lots. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements:

Requirement, operation or assembly	Rqm't para
Underplating of nickel on silver filled, rhodium plated insignia	3.8.2.2
Black chemical finish prior to enameling of subdued insignia	3.8.3

4.3.4 End item examination.- The defects found during examination shall be classified in accordance with 4.3.4.1, 4.3.4.2, 4.3.4.3 and 4.3.4.4. The lots shall be inspected in accordance with 4.3.4.5. The sample unit for the examinations in 4.3.4.1 and 4.3.4.2 shall be one insignia of each rank and size.

4.3.4.1 Examination of insignia for visual characteristics.- Examination shall be made at a distance of 16-18 inches with lumination equal to average

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daylight and arranged so as to avoid reflected light as much as possible. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance and serviceability and minor when affecting appearance and serviceability but not seriously.

Examine	Defects	Classification		
		Major	(*)	Minor
Front:				
Color and finish	Color or finish of metal does not compare favorably with standard or approved sample	x	-	-
	Color, shade, opacity or translucency of enamel (as applicable) does not compare favorably with standard or approved sample or specified color chip	x	-	-
	Enamel not type specified	x	-	-
	Enamel not uniform in color	-	*	-
	Discoloration such as spot, stain or speck	-	*	-
	Not clean or foreign matter imbedded	-	*	-
	Not plated, not type plating specified (when required)	x	-	-
	Any solder spatter exposed or concealed by plating	-	*	-
	Any solder spatter or discoloration on edge	-	*	-
	Not undercoated with black chemical finish (when required)	x	-	-
Enamel (hard)	Not enameled when required	x	-	-
	Enamel area cracked, crazed, pitted or chipped	x	-	-
	Enameled area contains lumps, bubble or blister	-	*	-
	Any fishscale, copperheads, scale or specking	-	*	-
	Enameled area contains a hole, extending through enamel to base metal	x	-	-
	Enameled area contains a sharp depression, such as a low place, not extending through to the base metal, affecting appearance	-	*	-
	Foreign inclusion in enamel, i.e., colored spot or speck amounting to a noticeable discoloration	-	*	-
	Scumming, lack of gloss or not polished	x	-	-
	Not level with dykes	-	-	x
	Any enamel not contained within prescribed outline of design	x	-	-
	Laps, overruns or skips	x	-	-

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Examine	Defects	Classification		
		Major	(*)	Minor
Enamel (baked)	Not enameled when required	X	-	-
	Laps, overruns or skips	X	-	-
	Foreign inclusion in enamel causing bumps, blister or discoloration	-	*	-
	Area on face or edge not enameled where required	-	*	-
Design	Construction or design details altered or obscured or not conforming to Government hub	X	-	-
	Any significant detail not clear, marred, missing, reduced or obliterated	X	-	-
	Any warp, twist or distortion, producing irregular surface contour or outline	X	-	-
	Any detail struck over resulting in double impression	X	-	-
	Any area not pierced as indicated by specification sheet	X	-	-
Quality of plating	Surface spotted or open grained, i.e., appearing as pitted, porous or crystallized	X	-	-
	Any plated area cut-through, not continuous, broken, peeled, blistered, shaded or otherwise impaired	X	-	-
Lacquering	Lacquer coating is not smooth or is not adherent, e.g., flaking, blistering or peeling	-	*	-
	Lacquer forms noticable runs	-	*	-
Workmanship	Surface or edge not clean, not smooth or free from burrs, drag, step or toolmarks	-	*	-
	Metal marks on exposed surface, such as nick, dent or scratch	-	-	X
	Die-struck edge not well defined	-	-	X
	Not trimmed or pierced as applicable, to the die-struck edge	-	*	-
	Pinhole, crack or rupture	-	*	-
	Design or shape altered by piercing	X	-	-
	Piercing not clean and smooth, e.g., any burrs, drag, cutter, drill or file marks	X	-	-
Back:				
Color and finish	Not plated when required	X	-	-
	Plating is not continuous, or is cut-through, porous or not adherent	X	-	-
	Not clean, i.e., discoloration, such as spot, stain or speck, over 3/16 inch in maximum dimension	-	X	-

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Examine	Defects	Classification		
		Major	(*)	Minor
	Not bright gold plated when required.....	-	-	x
Fastening device	Any part not as specified or wrong material.....	x	-	-
	Defective, i.e., any part damaged or malformed which affects use or which may be expected to affect use.....	x	-	-
	Deformed, i.e., any part misshapen, bent or damaged, but not affecting use.....	-	-	x
	Not positioned as specified, i.e., not conforming to specification sheet.....	-	*	-
	Any component omitted, misplaced, operation omitted or not properly performed.....	x	-	-
Workmanship	Impression of forcer when used not regular and symmetrical with the design.....	-	-	x
	Sharp or rough edges or corners in forced area, or thin fin of metal along trimmed or pierced edges.....	x	-	-
	Forcer used where not permitted.....	x	-	-
	Not straight flat back or curved back as applicable.....	-	-	x
Soldering	Component parts not completely joined by solder, any perceptible opening between component parts.	x	-	-
	Joints not clean and not smooth, e.g., flux or excess solder not removed.....	-	-	x
	Area burned or reduced in soldering.....	-	*	-
	Solder spatter exceeding 3/16 inch in maximum dimension.....	-	-	x
Contractor's identification and quality marking	Missing, illegible, misspelled, incorrect or not stamped.....	-	-	x
	Not inconspicuous on the back when viewed form the back.....	-	-	x
	Not located in the forced area when a mock forcer has been used.....	-	-	x

4.3.4.2 Examination of insignia for dimensions.- Any dimension that is not within the specified tolerances shall be classified as a defect.

4.3.4.3 Examination of insignia for weight.- The lot shall be rejected if the average weight is less than the minimum specified. The sample unit for this examination shall be 12 insignia of each rank and size, including attaching devices, but not clutches.

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4.3.4.4 Examination of Captain (Navy) or Colonel insignia for pairing.- Any Captain (Navy) or Colonel insignia mismated shall be classified as a defect. The sample unit for this examination shall be one pair of Captain (Navy) or Colonel insignia.

4.3.4.5 Examination of paired insignia for compatible finish.- Any paired insignia having finish and/or color which is not closely matched to each other shall be classified as a defect. The sample unit for this examination shall be one pair of insignia mounted on a card.

4.3.4.6 Inspection levels and acceptable quality levels (AQL's).- The inspection levels and the AQL's expressed in defects per hundred units shall be as follows:

	Inspection level	AQL	
	II	Major	Total
For defects applicable to 4.3.4.1	II	2.5	10.0
For defects applicable to 4.3.4.2	S-3	(one class)	4.0
For defects applicable to 4.3.4.3	S-1	(one class)	N/A
For defects applicable to 4.3.4.4	S-4	(one class)	1.5
For defects applicable to 4.3.4.5	S-4	(one class)	1.5

4.3.4.7 End item testing.- Testing of the completely fabricated insignia shall be performed in accordance with table III for the characteristics shown therein. The sample unit shall be three completely fabricated insignia for each rank and size. The requirements are applicable to the individual unit with one determination per sample unit, except all fused prongs on each sample unit shall be tested. The results shall be reported as pass or fail. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Table III End item testing.-

Characteristic	Rqm't Para	Test para
Test for soldered joints	3.6.2	4.4.7
Test for clutch fasteners and prongs	3.7.1.1	4.4.2
Test for fused prongs	3.6.2	4.4.3
Acid test for gold plating	3.8.2.1	4.4.5
Test for continuity of nickel and rhodium plating	3.8.2.2	4.4.6
Scratch test for baked enamel	3.8.3	4.4.4.1
Acetone test for baked enamel	3.8.3	4.4.4.2
Dye test for anodized surfaces	3.8.4	4.4.8

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4.3.5 Examination for count of insignia in intermediate containers.- Insignia packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (interior package). The lot size shall be the number of intermediate containers. Any box containing less than the specified or marked quantity of insignia shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.3.6 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking requirements as specified are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defects
Markings (exterior and interior)	Missing, incorrect, illegible or improper size, location, sequence or method of application.
Material	Any component missing. Any component damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight or content (exterior and interior)	Number of intermediate containers is more or less than required; net weight exceeds requirements.

4.4 Tests.-

4.4.1 Chemical composition.- Chemical composition shall be determined in accordance with the test method No. 111 of Fed. Std. 151. Results shall be evaluated to determine compliance with the requirements specified in 3.4.1.

4.4.2 Prongs and clutches.- Insignia shall have clutch fasteners removed and replaced ten times and checked for compliance with the requirements of 3.7.1.1.

4.4.3 Test for fused joints.- All fused prongs on each insignia shall be bent back and forth through an angle of 90 degrees until each prong breaks. The fused joint shall then be examined to determine compliance with 3.6.2.

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4.4.4 Test for baked enamel.-

4.4.4.1 Scratch test.- Draw a knife edge across the face side of the specimen making four parallel scores one-sixteenth inch apart. Repeat this process drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.8.3.

4.4.4.2 Acetone test.- The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton and enamel shall then be examined to determine compliance with 3.8.3.

4.4.5 Acid test for gold plating.- Insignia to be tested for compliance with 3.8.2.1, may be unlacquered or have the lacquer completely removed with an organic solvent. The acid test shall be conducted as follows: using a test acid consisting of a solution containing 50 percent by volume nitric acid (specific gravity 1.42) and equal volume of distilled water. At room temperature place a drop of test acid (each drop to be 1/8 inch minimum diameter) on three different spots on face of insignia. The drops shall remain on the insignia for not less than two minutes. A minimum of two of these spots shall withstand this test.

4.4.6 Continuity of nickel and rhodium plating.- The rhodium plated insignia shall be immersed for three minutes in a solution of liver of sulphur (2 percent potassium sulfide and 98 percent water by weight) at 100 degrees F. plus or minus 10 degrees F. After removal, insignia shall be washed clean in water and examined for compliance with 3.8.2.2.

4.4.7 Soldered joint test.- Samples to be tested shall be placed in an oven maintained at 1055 degrees F., plus or minus 5 degrees F., for not less than 15 minutes. While at the specified oven temperature, samples shall be lifted by the smallest part of the soldered component to determine compliance with 3.6.2.

4.4.8 Dye stain test for anodized surfaces (ASTM B-136-63T).- A spot of phosphate-free anthrogumone violet RN dye shall be placed on the anodized surface. After five minutes the dye shall be wiped off and the surface examined to determine compliance with 3.8.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (See 6.2).

5.1.1 Level A.-

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5.1.1.1 Unit package.- One matching pair of insignia of one rank and size only, shall be mounted on a white sulfite board card having a minimum thickness of 0.011 inch and measuring approximately 3-1/2 by 2 inches. Each mounted pair of insignia shall be individually packaged in a snug-fitting transparent polyethylene bag. The polyethylene flexible film shall be a minimum of 0.002 inch thick. All seams and closures shall be effected by heat sealing. Prior to or during the heat sealing closure operation excess air within the bag shall be expelled.

5.1.1.2 Intermediate package.- One hundred pair of insignia of one size and rank only, prepared as specified in 5.1.1.1, with insignia fronts facing each other in a snug-fitting set-up paperboard box donforming to type I, variety 1, class A, style 4 of PPP-B-676. Box closure shall be in accordance with the appendix of the container specification.

5.1.2 Level C.- Insignia shall be packaged to afford adequate protection against deterioration and damage during shipment from supply source to the first receiving station. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Insignia of one type and size only packaged as specified in 5.1, shall be pack in a snug-fitting shipping container conforming to class 2 of PPP-B-621 or overseas type style A, B, I, or J of PPP-B-601. Closure and reinforcement shall be in accordance with the appendix of the applicable specification.

5.2.2 Level B.- Insignia of one type and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberbaord shipping container conforming to class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.2.3 Level C.- Insignia, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, interior packages and shipping containers shall be marked in accordance with MIL-STD-129, except unit package marking may be printed on the reverse of the mounting card provided the printing is continuous and legible through the bag.

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6. NOTES

6.1 Intended use.- Rank insignia are intended to be worn on uniforms to identify the Officer's rank or grade.

6.2 Ordering data.- Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.-

(a) Title, number and date of this specification.
(b) Title, number and date of applicable specification sheet (see 1.2 and 3.1).
(c) When a standard sample will not be furnished (see 3.2).
(d) When a first article is not required (see 3.3).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.2 Contract data requirements.- Data conforming to Department of Defense Data Item Description DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps procurement, the cognizant procuring activity will provide comparable Data Item Descriptions. When so required, such data will be specified for delivery on Form DD-1423 included in the contract.

6.3 Aluminum alloys.- While any suitable aluminum alloy may be used for anodize, it should be noted alloys in the 1100 series have been recommended for silver color and subdued insignia and alloy 5557-5567-H25 temper or alloy 6463-T42 temper have been recommended for gold color insignia. The aforementioned alloys have been recommended as suitable for the finishing, anodizing and drying process necessary to attain the color required.

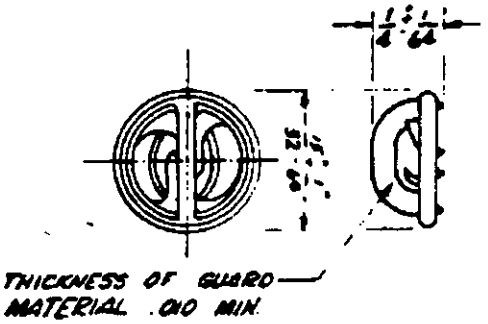
Custodians:
Army - IH
Navy - PE
Air Force - 11
Review Activity:
Navy - PE, MC
Air Force - 11
Other - CT
User Activity:
Army - SM

Project No. 8455-0138

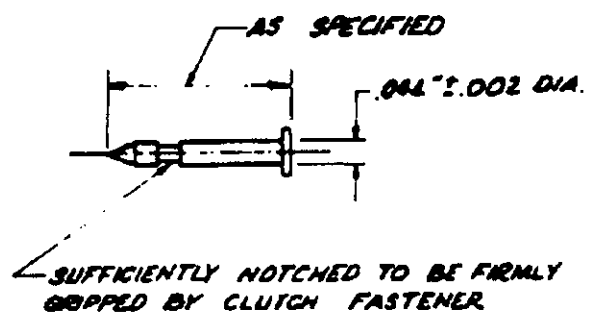
Preparing Activity:

The Institute of Heraldry,
United States Army

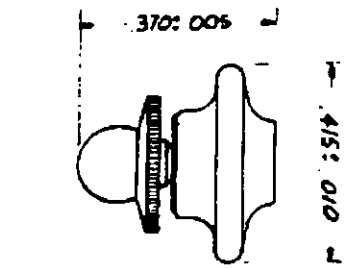
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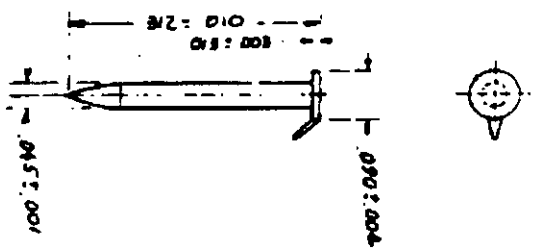
CLUTCH



PRONG



BALL TYPE CLUTCH



PRONG WITH PICK

This is a miniature copy of
THE INSTITUTE OF HERALDRY
Dwg No. B-13-12 dtd 14 Dec 1970
This copy if for information
only, the dwg. referenced in
paragraph 2.1 governs.

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-I-3949D Insignia, Rank, Officer's, Metal; Gen Spec For

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 1426

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